CATERPILLAR DEALER BEST PRACTICE

Inversion of the body pivot pins on mining trucks

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The removal and installation services of truck body assembly is considered a highly critical activity, due to the inherent risks, from the load motion activities and the high weight of the parts involved.

One of the task stages is the removal and installation of the pivot pin from the body. The project of the Off-highway truck 793D has the pin fixation positioned to the external part of the equipment, as can be seen in the Figures below.

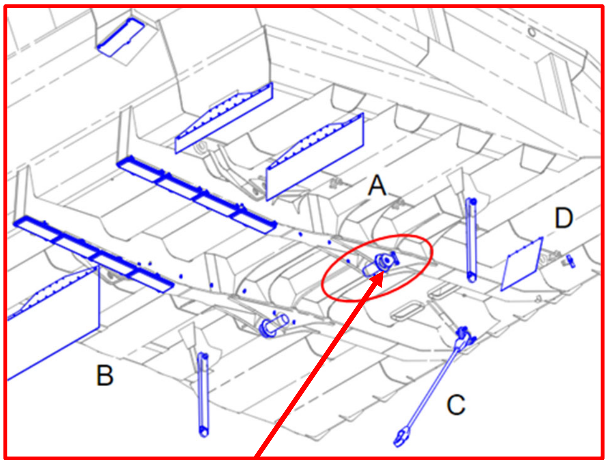

Figure 01: picture from SIS relative to the positioning of the body pivot pins, acooring to factory assembly 793D.

Figure 02: picture from the positioning of the body pivot pins, acooring to factory assembly.

If there is no working at heights package installed, this best practice has as objective show an alternative of repositioning of the pin from the body, reducing the technician exhibition in the work at height and improving the ergonomics.

2.0 Best Practices Description

The locking pin for the pivot pins of the body 793D is on the outside of the chassis, during the removal or installation of the pin, the technician needs on the tire, in order to support himself, he cannot use a seat belt to work in height to have access to the pin, bringing a high risk to the activity due to inadequate ergonomics, knowing that the weight of the pin is 40kg

In order to reduce the risks inherent in the activity, the body pivot pins were reversed, where we started to install the body pivot pins from the inside of the chassis, as shown in figures 03 and 04. It is recommended to position the square blocks exactly where the exterior ones are located with respect to the pivot pin access.

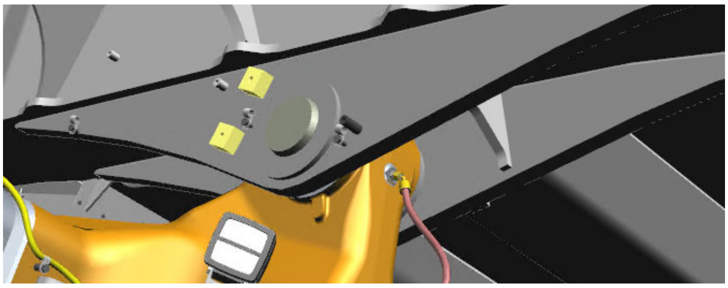

Figure 03: Picture of the pin lock in the internal part of the stringer rail body.

Figure 04: Picture of the pin positioning without lock to the external part of the stringer rail body.

With this new positioning of the pivot pins of the body, the technician gains access to the pin with the platform utilization, without the need of neither wearing a working at heights harness or leaning on the tire. The figure 05shows the platform model used to execute the activity, allowing the pins removal and installation without necessity of seat belt utilization and with adequate ergonomic.

Figure 05: Platform to realize the activity pivot pins of the body installation and removal.

3.0 Implementation Steps

The improvement, inversion of the pivot pins body, can be realized during the gap in the preventive maintenance of body and canopy replacement. According to the figure 06, install the pins in an inverted Weld two boss (PN 8X-3097) to fix the plate 8W-8677, to lock the pivot pin. follow this step to both two of the stringer rail body.

This figure shows the blocks welded higher than desired. It’s important that the welds are at least 50mm away from the edge of the top welds. For best fitment, the blocks should be mirrored from the outside facing blocks.

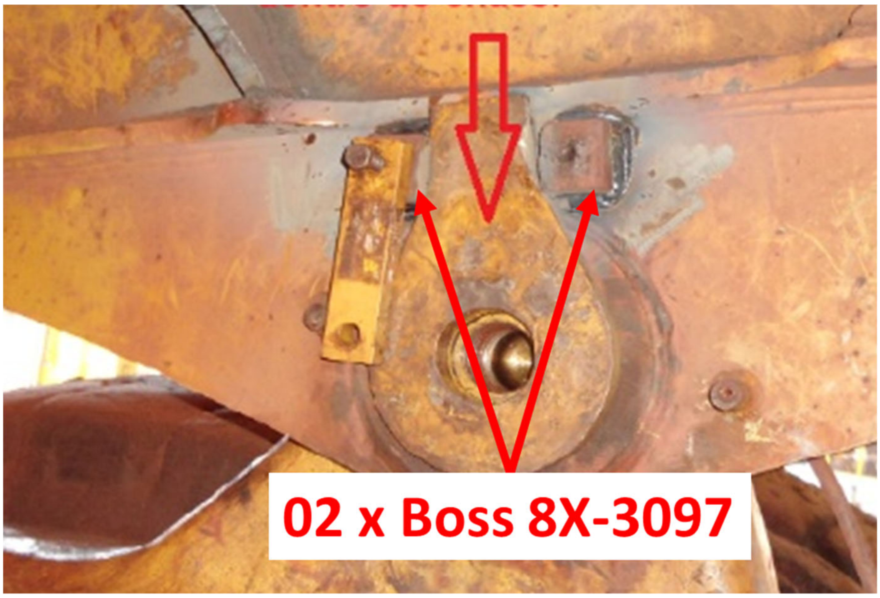

Figure 06: positioning the boos to the pin lock.

4.0 Benefits

- Could reduce the risk accidents during assembly and disassembly, while also reducing technicians' effort to install and remove the pins.
- Better ergonomics and mobility to realize the assignment, without the need of wear the seat belt.
- Reduce the activity time of dump removal or installation.

5.0 Resources Required

To performance the activity during the body replacement, it is necessary add the following resources bellow in the schedule:

- 04 supports PN 8X-3097.
- 01 working hour of the welder.

If the activity be done out of the body replacement activity, it will be necessary to add the resources to remove the pin.

6.0 Supporting Attachments / References

Illustrations and references cited in best practice.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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